

Observational Studies

College of Charleston

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Data Collection Methods

Broadly there are two major ways to collect data

Experiments where the variables of interest are manipulated (hopefully intelligently)

- ▶ Handing a subject a bean or letting subject choose the bean
 - ▶ Assigning different fields different new soybean hybrids
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Observational studies is where the researcher does not intervene in/manipulate the scientific process

- ▶ Phone screen time generally and on their favorite app (retrospectively)
- ▶ Asking someone how long they can plank for and then recording their time (prospective)

Observational Studies

Broadly two types...

- ▶ Retrospective studies
 - ▶ Looks backwards
 - ▶ Usually considered “weaker”
 - ▶ Asking about nutrition as a child for studying adult heights

- ▶ Prospective studies
 - ▶ Looks forward
 - ▶ Continues to take measurements into the future
 - ▶ Stronger than retrospective studies but weaker than experiments
 - ▶ Medical studies where they follow people around for years....

Causation: Right out

No control over variables and random assignment; it just **is**

Better to talk about correlation

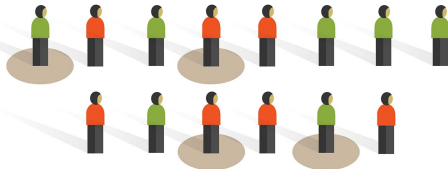
(Applies to experiments as well)

Survey Methods (the good ones)

Simple Random Sample: All population members have an equal chance of being chosen

- ▶ Most basic form
- ▶ Eg Grinnell sends an email survey to a random sample of students on LGBT acceptance/discrimination

Simple random sampling

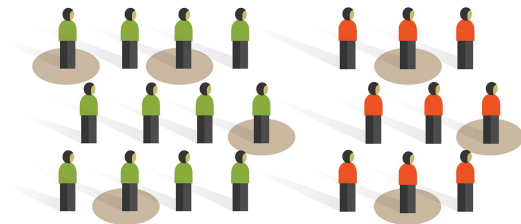


Survey Methods (the good ones)

Stratified Sampling: Simple random sampling occurs within each subpopulation

- ▶ The population is broken into groups and then each group gets sampled
- ▶ Members of a subpopulation of the same probability of being picked
- ▶ Members do not have the same probability of being picked across/between subpopulations
- ▶ Eg CofC sends an email survey on LGBT acceptance/discrimination to..
 - ▶ LGB Students
 - ▶ Trans/NB Students
 - ▶ Het-Cis Students
- ▶ Ensures all subpops are represented

Stratified sampling

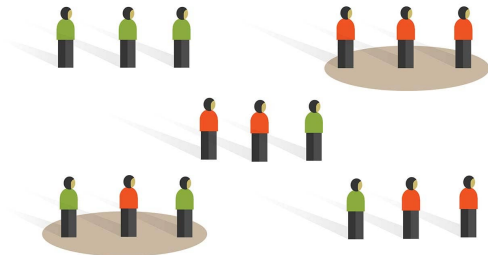


Survey Methods (the good ones)

Cluster Sampling is where all members in a few random (usually tiny) subpopulation are studied

- ▶ Exhaustive study of a few small things
- ▶ Eg CofC surveys by physical paper 5 randomly selected classes, all students required to respond

Cluster sampling



Survey Methods (the good ones)

Multi-Stage Sampling uses combinations of the others to make it easier, done in stages

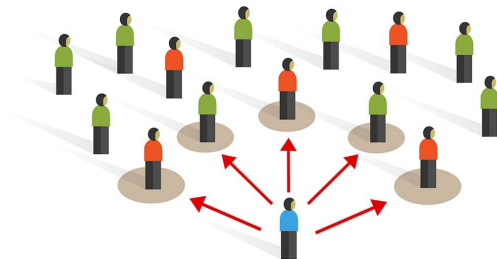
- ▶ Common for complicated situations
- ▶ Tedious?
- ▶ Eg CofC randomly samples 3 classes from each of the Humanities, STEM, Social Sciences. All students required to respond.
- ▶ Stratified (Humanities, STEM, Social Sciences)
- ▶ Clustered (entire classes)

Survey Methods (the bad ones)

Convenience Sampling is where a sample is taken because it is convenient/easy

- ▶ Eg I texted my friends and we all agreed that X
- ▶ **Undercoverage** is where members of the population cannot be sampled or are sampled disproportionately low
- ▶ Eg Asking people how long they can plank *only* at the gym doesn't give you a sense for the student body as a whole (pun)

Convenience sampling

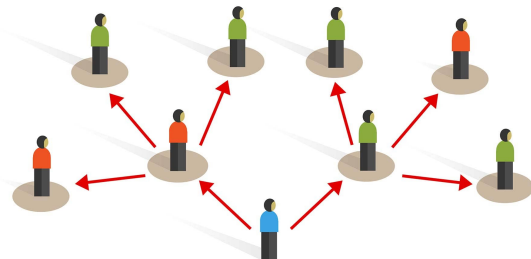


Survey Methods (the bad ones)

Snowball Sampling is where we use the first subjects to find the next few subjects

- ▶ Often useful (unfortunately) when populations are hard to reach/interact with
- ▶ Examples were early studies of homosexuals
- ▶ Problem is independence between respondents....

Snowball sampling



Several things can cause biases in your sample

▶ **Wording of the Questions**

- ▶ wording is so charged people are “pushed” one way or the other
- ▶ Also applies if it's so complicated/technical/run on as to be confusing

▶ **Response Bias**

- ▶ Why did this weirdo take time out of their day to fill out my survey?

▶ **Non-Response Bias**

- ▶ Why did this weirdo not take time out of their day to fill out my survey?

Do's and Don't's

Biggest issue with sampling is talking about who you were **actually** able to sample from. That becomes your de facto population even if you wanted to say something more broadly.

- ▶ Eg if you sampled only CofC students you can't say anything beyond CofC
- ▶ Eg A study of crapie in the upper end of the Mississippi River can only talk about crapie populations in the upper end of the Mississippi River
- ▶ Once the more limited population is identified it's easier to avoid undercoverage bias